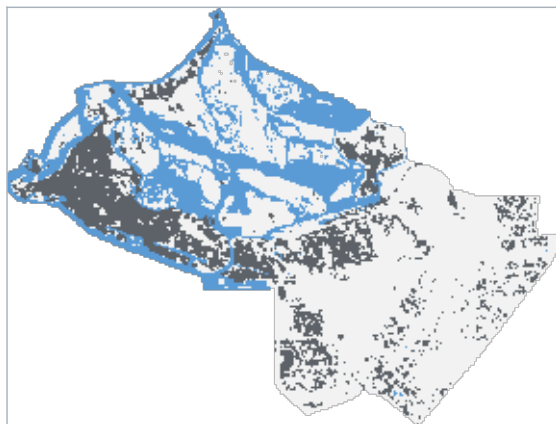
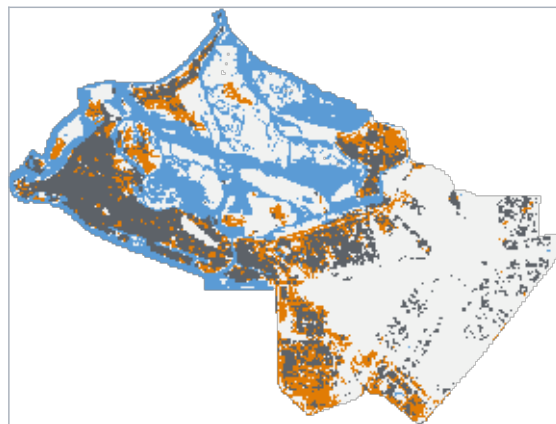


High-outward-growth scenario: model-specific 2031 built-change footprints

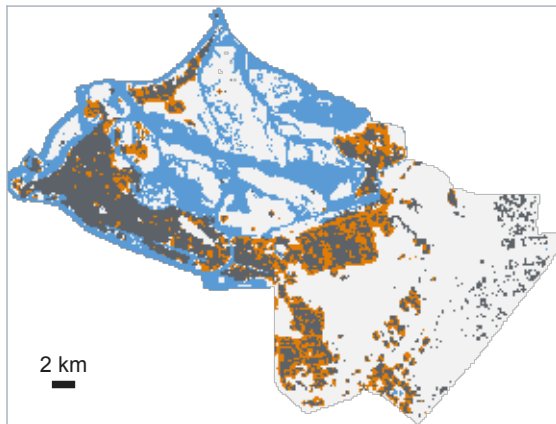
A Observed 2024 reference



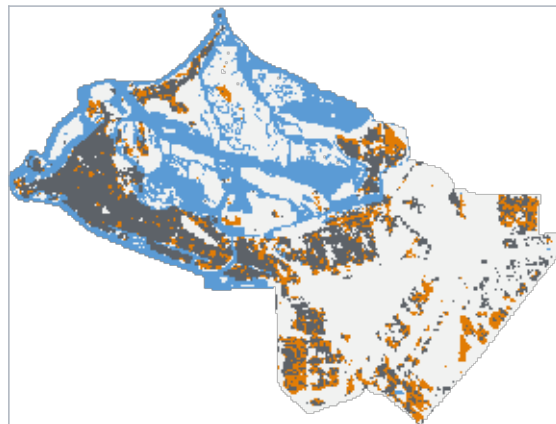
B GeoSOS-derived FLUS-style ANN-CA | 2031



C Geospatial Kernel | 2031



D GeoFM-LDN | 2031



Each map is a 100-m majority-vote ensemble. The full 3 scenarios x 3 models atlas is supplied as Fig. S1; change footprints are raster-cell products and not cadastral parcels.

Other valid land cover ■ Water or wetland ■ Built in 2024 ■ New built, 2024-2031